

Title: Multivariate Direct Filter Analysis

Subtitle: About Prediction Problems and Research Priorities in Forecasting

1. Blurb

This book examines forecasting within the framework of multivariate time series. It introduces a novel methodology—Multivariate Direct Filter Analysis (MDFA)—that emphasizes the structure of prediction problems and the importance of aligning forecasting methods with the forecaster’s research priorities.

Prediction tasks may take several forms. One may seek to forecast the next observation (one-step-ahead), predict values further into the future (multi-step-ahead), or estimate weighted combinations of future observations. The latter arises naturally in applications such as signal extraction, trend estimation, and seasonal adjustment. Because these objectives differ in structure, no single fixed criterion can adequately address them all.

Forecasting also involves several partly competing goals: accuracy in predicting future levels, timeliness in avoiding delays or excessive anticipation, and smoothness in suppressing spurious noise. These goals form what we term the ATS trilemma: improvements in any one of the three components—Accuracy, Timeliness, and Smoothness—inevitably come at the expense of at least one of the others.

The MDFA integrates these dimensions within a unified optimization framework. The criterion is tailored to the structure of the prediction problem while explicitly incorporating the ATS trade-offs that characterize forecasting practice. The approach is highly general: classical linear forecasting methods can be reproduced as special cases and subsequently refined to reflect specific research priorities. It is also interpretable in the sense that the optimization criteria follow simple, fundamental principles, yielding closed-form and uniquely determined solutions. These properties are particularly valuable in settings where black-box methods are either prohibited — as in applications subject to compliance or regulatory requirements — or simply undesirable, such as when the user seeks a deeper understanding of the forecasting mechanism.

Although the book provides a rigorous mathematical treatment, it also emphasizes practical implementation. Extensive applications, based on both simulated and real-world data, are presented as exercises using an open-source MDFA R package. These examples illustrate the methodology across a multiplicity of prediction problems and benchmark its performance against established approaches.

2. Rationale and Scope

To provide a deeper insight into your project, please talk around these points:

- What is motivating you to write this book? Why is now the right time?
- What are the essential prerequisites that you are assuming of readers? What will your book equip readers to do?
- Why have you taken the approach you have? Why have you chosen to include/exclude the things you have?

Motivation for MDFA

At the Institute of Data Science, 50% of financing is secured through research projects conducted in partnership with economic actors, industry stakeholders, and economic institutions. These projects emphasize real-world challenges — such as managing effective forecast uncertainty when predicting healthcare costs or GDP multiple quarters ahead or future asset prices — and place strong demands on both outperformance relative to established benchmarks and interpretability. Black-box approaches, including machine learning methods, are typically excluded either due to the project partners' priorities or external constraints such as the prevailing regulatory environment.

Achieving systematic, i.e., explainable outperformance requires that the structure of the prediction problem be translated into transparent optimization criteria, while explicitly reflecting the priorities of the research partner. Different applications naturally emphasize different aspects of forecasting performance. A GDP nowcast, for example, may place primary weight on accurately estimating the current level of economic activity. By contrast, a forecast further into the future may prioritize effective lead time and early detection of turning points. At the same time, the integration of noisy high-frequency indicators—often crucial for timely nowcasting—requires methods that place greater emphasis on suppressing spurious fluctuations and extracting stable signals from volatile data.

The MDFA is uniquely suited to address these challenges. It accommodates general prediction problems, is fully interpretable, and is numerically tractable, offering fast, closed-form, and uniquely determined solutions. Furthermore, it can replace, complement, or extend existing tools to meet specific user needs. Beyond the inherent merit of introducing a novel and generalizing framework, we are particularly motivated by the growing demand for interpretable, computationally efficient methods grounded in new theoretical insights — methods that directly and transparently address the known shortcomings of existing approaches and benchmarks.

Prerequisites and Payoff

We assume readers of this book to be fluent in classical time series analysis. Our aim is to equip them with the ability to identify the shortcomings of classical forecasting methods, understand the structure of the most common prediction problems, formulate their own

priorities, replicate and enhance existing benchmarks, and resolve these issues within a consistent methodological framework. For the more theoretically inclined reader, the book provides new insights into the non-stationarity of time series and into forecast optimization principles that generalize classical approaches, supported by rigorous mathematical treatment. For the more practically oriented reader, comprehensive exercises illuminate the practical facets of the theoretical concepts, benchmarking performance against established references.

The Whys

Our approach is built on identifying and addressing the shortcomings of widely accepted and commonly implemented benchmarks in forecasting practice. We propose a rigorous theoretical treatment that grounds the MDFA in a formal framework based on simple, transparent principles, extending classical approaches in directions that are both relevant to practitioners and of interest to theoreticians. Certain aspects that have not yet been sufficiently explored in applied research have been deliberately excluded, but may become relevant in a subsequent updated edition of the book.